

PUBLIC SUBMISSION

As of: March 21, 2025
Received: March 14, 2025
Status: [REDACTED]
Tracking No. m89-e98z-g470
Comments Due: March 15, 2025
Submission Type: API

Docket: NSF_FRDOC_0001

Recently Posted NSF Rules and Notices.

Comment On: NSF_FRDOC_0001-3479

Request for Information: Development of an Artificial Intelligence Action Plan

Document: NSF_FRDOC_0001-DRAFT-1542

Comment on FR Doc # 2025-02305

Submitter Information

Email: [REDACTED]

Organization: Intelligent Transportation Society of America

General Comment

The Intelligent Transportation Society of America (ITS America) respectfully submits comments on the Request for Information on the Development of an Artificial Intelligence (AI) Action Plan from the National Science Foundation and Office of Science and Technology Policy. ITS America is grateful for the opportunity to provide feedback to the Trump Administration on how AI already is transforming the transportation industry, in addition to how it can help shape the future. Our response will also discuss the role that Federal leadership should play in guiding both public and private sector innovation in this sector. In this response, we have provided recommended policy actions and real-world case studies of AI in transportation in the appendix. Please find our response attached.

Attachments

OSTP AI Action Plan ITS America Response_Final

OSTP AI Action Plan ITS America Response

March 14, 2025

Mr. Kirk Dohne
Acting Director
Networking and Information Technology Research and Development
National Coordination Office

Re: Request for Information on the Development of an Artificial Intelligence (AI) Action Plan; [FR Doc. 2025-02305]. This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

Introduction

The Intelligent Transportation Society of America (ITS America) respectfully submits comments on the Request for Information on the Development of an Artificial Intelligence (AI) Action Plan from the National Science Foundation and Office of Science and Technology Policy (OSTP). ITS America is grateful for the opportunity to provide feedback to the Trump Administration on how AI already is transforming the transportation industry, in addition to how it can help shape the future. Our response will also discuss the role that Federal leadership should play in guiding both public and private sector innovation in this sector. In this response, we have provided recommended policy actions and real-world case studies of AI in transportation in the appendix.

ITS America is the only national organization bringing together the private and public sectors in deploying intelligent transportation. ITS America convenes experts across the industry fostering innovation in intelligent transportation systems. Our member organizations include state, regional, and local transportation and planning agencies, transit providers, companies providing intelligent transportation system products and services, major technology firms, auto manufacturers and suppliers, research organizations, academic institutions, and other industry practitioners to deliver efficient, effective technology-forward transportation solutions.

AI is being used today all across our nation's transportation system, including freight and logistics, passenger vehicles, public transit, rail, aviation, and even maritime. The public and private sectors are working together to deploy AI-driven tools today to detect pedestrians in crosswalks, optimize traffic flow to reduce congestion, proactively manage physical infrastructure assets, hold intersecting lights when vehicles are entering an intersection dangerously and running a red light, and streamline internal public agency operations to maximize cost savings. Nations around the world, including China, have already invested billions in technological innovation, including AI, into their transportation system. The Trump Administration has a unique opportunity to establish the U.S. as the uncontested leader in transportation innovation with AI so that we remain globally competitive in this critical sector.

Transforming How We Move and Build: Artificial Intelligence in Transportation

We have seen how other sectors, such as healthcare, financial services, and defense, are utilizing AI – now it is

time to bring such attention to our transportation system.

Robust and scaled investments in transportation technologies, including AI, can enable a transformation of our transportation system to one that prioritizes safety and efficiency. Technology-enabled interventions afford us the opportunity to proactively prevent fatalities and injuries on American roads, allowing us to begin to meaningfully reduce the 40,000 lives on our roadways each year. Technology solutions allow us to create a more efficient transportation system and give people time back to spend with friends and loved ones, rather than continuing with a status quo approach which has resulted in Americans losing 54 hours to congestion while costing our economy an estimated \$224 billion annually. AI-enabled transportation solutions can deliver a truly modern infrastructure network which rises to meet the needs of our communities, making transportation safer and more efficient, enhancing the longevity of our transportation resources, and bolstering our readiness for severe weather events – readiness which cannot fully be similarly reached without these tools.

Our transportation system is no longer just concrete, asphalt, or steel – it is a combination of the physical and digital layers, increasing our ability to better plan, build, operate and maintain our system by utilizing the intelligence coming from the digital layer. Today, there are many disparate data sources contributing to transportation operations – both historical and real-time - which can be used to inform safety solutions, streamline operations, improve asset management and more. AI is a critical tool to leveraging these data sources and turning the data into actionable insights to improve outcomes for all who use our system. With the right Federal leadership and investment, we can use AI to harness this data at scale.

We have already seen some Federal leadership on AI from bodies like the National Institute of Standards and Technology (NIST) and the AI Safety Institute, both of which play an important role in testing models and ensuring U.S. technical leadership. The U.S. Department of Transportation (USDOT) has begun funding projects that utilize AI through initiatives such as the Strengthening Mobility and Revolutionizing Transportation (SMART) Grant program, the Intersection Safety Challenge, Complete Streets & AI Initiative for small businesses, the Advanced Research Projects Agency – Infrastructure (ARPA-I), and the ITS Joint Programs Office. USDOT has an opportunity, with the full support of OSTP, to strengthen and build upon these efforts, encouraging further deployment of innovative technologies that enhance safety and mobility.

By fully leveraging AI, we can enable a future where congestion is reduced by optimizing a network of traffic signals with the click of a button getting people home faster to their loved ones; a future where we know about structural issues on a bridge long before a tragedy occurs; a future where truck parking becomes as simple as making a reservations a restaurant; a future where traffic signals detects a pedestrian needs more time to cross a busy road and adjust in real time; a future with fewer vehicle crashes; and a future where the U.S. leads the world in transportation innovation. OSTP and the Trump Administration have a great opportunity to use the AI Action Plan to envision a bold future where technologies like AI are seamlessly integrated throughout all our transportation and infrastructure processes, and policies are designed in a way that will allow us to take full advantage of AI's benefits.

Below you will find our core policy action recommendations for the AI Action Plan.

AI Policy Action Recommendations

I. Deploying AI Through USDOT Programs

USDOT should encourage State and local agencies who receive Federal funding to deploy technology, such as AI, to improve safety and efficiency outcomes. There is a significant opportunity for Federal leadership in funding and incorporating technology like AI in USDOT programs, including formula programs, discretionary grants, workforce development initiatives, research and development, and transportation planning. The AI Action Plan should promote further integration of widespread technology use in Federally funded transportation projects.

This will ensure that the U.S. remains a leader in transportation innovation while working to improve outcomes in communities across the country. Given its ability to analyze vast amounts of data, recognize patterns, and deliver key insights, AI can make transportation operations more efficient, saving agencies money and time. As the country looks to remain globally competitive for years to come, it is important that USDOT take a leading role in encouraging technology adoption.

The amount of funding currently directed towards transportation technology is inadequate relative to the positive impact such funding can create, and we must move beyond research and development into sustained, widespread deployment. We have already seen strong commitments from the Trump Administration on AI investments and technology in the U.S., and this should apply to the transportation sector as well. The Administration has an opportunity to lay a marker down on a bold vision for the future of American transportation by prioritizing broader technology and AI adoption to improve safety and efficiency outcomes.

As with many emerging technologies, targeted Federal involvement will accelerate the pace of change and speed of deployment for a digital infrastructure marketplace that is a platform for innovative technologies. This digital infrastructure will allow us to build longer-lasting transportation infrastructure more efficiently and solve challenges related to safety, congestion, and our national supply chain. Current Federal transportation initiatives, as well as many State and local initiatives, do not yet include adequate consideration for technology and digital infrastructure. Technology elements can be mistakenly considered as an “add-on” or “nice-to-have” – but in this era of AI at our fingertips, transportation agencies across the country need to modernize and consider digital tools in a more fulsome way to reduce roadway crashes and improve the overall travel experience.

The Federal government should continue robust engagement with stakeholders. The public sector, private sector, and academia should all be at the table when formulating policy and plans for the use of AI. The Federal government, including Congress and the Executive branch, should continue robust engagement and dialogue with stakeholders of all backgrounds to ground AI policy in safety and tangible, beneficial outcomes. Buy-in is needed from a broad set of stakeholders to proliferate AI applications and adoption, and we need further engagement on practitioner education, lessons learned, and research dissemination.

II. Ensuring Data Privacy and Cybersecurity

ITS America calls for Federal leadership to create a national data privacy standard. This would mitigate uncertainty across the country and the patchwork of State laws which are inhibiting economic growth and broader innovation in the transportation sector. As AI technologies continue to evolve and spread across a variety of industries and geographies, a comprehensive, national standard should be enacted. Data privacy protections ensure that the public’s information is not being collected without consent. Without this, adoption

and public acceptance of AI applications in transportation, among other industries, will be hindered. In addition, the AI Action Plan should include a focus on data governance. A data governance framework from the Federal government would give both the public and private sector guidance for ensuring collected and processed data is of high quality and protected from unsafe actors, while mitigating organizational risks.

As part of the proposed AI Action Plan, all levels of government and industry, including USDOT and transportation sector, should adhere to the most recent NIST standards, currently the AI Risk Management Framework and Cybersecurity 2.0 Framework.¹ The U.S. government must prioritize a secure AI ecosystem that includes robust cybersecurity and data privacy measures. AI-related transportation assets rely on various interconnected systems, such as sensors, communications methods, and data processing systems. This greater degree of connectivity can create additional opportunities for hostile actors to access critical transportation infrastructure, threatening to cause significant disruption to the traveling public. Cybersecurity considerations should remain a top priority during the design, development, and deployment of AI tools.

Further, ITS America recommends that policymakers dedicate additional support for building greater professional cybersecurity workforce capacity within the transportation sector in order that transportation agencies have a sufficient workforce capable of preventing and responding to cybersecurity incidents.

III. Addressing AI for Automated Vehicle (AV) Technology

The AI Action Plan should consider AVs and advanced driving technology that improves road user safety and access to mobility options. The AI Action Plan from OSTP should call for the creation of such a framework and should call on USDOT to clarify and codify interpretations to ensure industry certainty on technologies and human interaction with AVs. AVs hold tremendous promise for improving roadway safety and increasing mobility options for communities of all kinds. Along with ADS Level 4 and 5, Advanced Driver Assistance Systems (ADAS) have the potential to significantly reduce crashes on our roadways. These vehicles rely on AI-powered algorithms and decision-making on the road. ITS America supports policies and regulatory frameworks that facilitate the safe testing, deployment, and integration of these technologies into the transportation system while simultaneously developing strategies to support our existing and future workforce.

In addition, **the National Highway Traffic Safety Administration (NHTSA) and Federal Motor Carrier Safety Administration (FMSCA) should finalize a comprehensive AV safety framework** expeditiously so that we can fully realize the benefits of automation on our roads on a national level.

IV. Promoting Explainability, Transparency, and Trust

The OSTP AI Action Plan should direct USDOT to develop transparency guidelines for AI applications in coordination with the industry. Bringing AI innovation to transportation, a sector where safety and consumer acceptance is paramount, will require AI systems to be explainable, transparent, and trustworthy. Without transparency, advanced AI can be susceptible to unethical uses and reduce public trust. Establishing a

¹ ¹ National Institute of Standards and Technology. <https://www.nist.gov/itl/ai-risk-management-framework>

level of transparency in data collection, algorithms, and how AI applications are used will help ensure trust among users. Transparency about data and AI models encourages responsibility, decreases the risk of misuse, and will ultimately lead to better tools and better results for the American public. USDOT and other Federal stakeholders should develop robust reporting requirements for any harmful incidents associated with AI utilization in the name of greater transparency.

Additionally, **ITS America recommends the development of explainability standards in the Action Plan for the use of AI systems in transportation, as well as other industries.** AI tools should be explainable to those without expertise in technology or a specific industry. Without strong explainability, it will be more difficult to improve consumer adoption of technologies, will further erode trust, and may be prohibitive to future private sector investment. If there is a lack of trust among consumers, private companies may not be incentivized to innovate and create new products and services in the AI realm.

The Administration's Action Plan should ensure that the AI Safety Institute continues its work in creating safety standards, testing new technologies, and evaluating AI models. Without such a body, the U.S. risks ceding AI leadership to non-allied countries such as China and opening the door to the proliferation of unsafe and untrustworthy AI models. The AI Safety Institute has broad industry support beyond the transportation sector, as well as bipartisan support from Congress.

V. Strengthening Workforce Development and Education

ITS America recommends that the Administration, in coordination with Federal agencies such as the Department of Labor and USDOT, develop a robust workforce development plan and assistance program to ensure our transportation workforce is prepared to leverage AI tools and solutions. In the era of AI, workforce development and education are critical to fostering broad adoption and ensuring a pipeline of innovation. In any industry, upskilling and reskilling may be necessary to fully maximize the benefits of AI applications. This includes training the existing workforce on how to leverage AI tools and solutions to support their own work, in addition to AI's capabilities and limitations. AI should be used for decision support and augmentation of tasks, not as a replacement, and workforce development programs should reflect that.

In the AI Action Plan, OSTP should place emphasis on the need to conduct AI training for practitioners and those who will be interacting with and using AI on the job. Federal leaders can and should put the transportation workforce ahead of the curve on AI. As AI adoption grows in the transportation industry, we must ensure that Federal transportation workforce programs are prioritized and well-funded for digital skills training.

For future workforce needs, OSTP and USDOT should continue to support the development of AI training curriculum related to transportation at universities and should prioritize AI as a key component of the University Transportation Centers program. Additionally, AI literacy and curriculum should be a core component of community colleges, apprenticeships, career and technical education, and even primary and secondary education as we look to prepare our workforce for the future.

VI. Modernizing Procurement Practices

The AI Action Plan should discuss ways to streamline the contracting process for technology solutions, such as AI, and OSTP should work with USDOT to provide practices for procuring AI-based tools for transportation projects. This may include continued research and pilots on innovative procurement methods through ARPA-I or FHWA's Every Day Counts program. Procurement remains a core barrier to AI adoption in transportation. Procurement policies designed for physical infrastructure do not adequately consider the procurement, maintenance, and operations needs of a technology-inclusive infrastructure system. Data streams, software licenses, and even cloud storage services are key to the transportation system today and the use of AI but were not core parts of our infrastructure procurement thirty years ago. AI solutions are distinctly unique from physical assets – they are not one-time purchases, and they often require ongoing upgrades to ensure cybersecurity and data integrity and can improve performance over time unlike a static physical asset.

Procurement on Federally-funded transportation projects should allow for and encourage license-based or subscription products that are often associated with technologies like AI – these are constantly updating assets and not one-time hard purchases. The current transportation technology procurement process can be lengthy and onerous, especially for States and localities that may lack the necessary resources and expertise, slowing down projects and threatening the efficiency of investment in technology solutions. Additionally, Federal regulations pertaining to operational costs in Federal-aid Highway projects make it difficult to fund the ongoing operations and maintenance of assets. AI assets require ongoing maintenance and optimization to work effectively.

Conclusion

To solve our safety crisis on the road, make travel more efficient, and spur economic growth, we must take bold steps toward transforming how we integrate transportation technologies, including AI, into our transportation system at scale. AI capabilities are advancing at a fast pace, and we need a Federal policy environment that allows us to more holistically integrate these tools today so that we are prepared to take advantage of the new and improved capabilities of tomorrow. Doing things the old way has resulted in less safe roadways, more congested commutes, slower movements of goods and broken supply chains, and decreasing global competitiveness. Now is the time to accelerate our investments in AI and empower our transportation system for a better tomorrow. We look forward to seeing significant consideration of transportation in the Administration's AI Action Plan.

If you have any questions, please contact Bobby McCurdy, Vice President of Policy & Advocacy at

Laura Chace
President and CEO
Intelligent Transportation Society of America

Appendix: Applications

a) Proactive Maintenance (Infrastructure and Vehicles) and Asset Management

Using AI, transportation agencies can predict maintenance needs proactively for infrastructure assets and fleets. AI tools can help agencies identify structural issues in bridges and roads early on and monitor for further wear-and-tear. Traditional methods involve manual inspection, which can be costly and time consuming for public agencies. New innovations use computer vision AI, virtual reality tools, and drones to collect images and videos of roads and bridges. While the cameras continually monitor the site, software processes and analyzes the collected data, providing engineers with a safety assessment that includes information about structural changes and weaknesses, as well as immediate damage.

Using machine learning and other AI tools, automakers and manufacturers can now identify and predict vehicle and fleet maintenance to a more precise degree, improving not only vehicle performance but also maintenance operations and costs. New York City's MTA is using AI technology to predict bus breakdowns and maintenance needs, helping the agency plan for future needs and operate efficiently for customers. MTA found a 75% increase in maintenance productivity and material costs lowered by 24%.

Maintenance through software-enabled assets is a key area where USDOT and the AI Action Plan could focus, bringing more cutting-edge AI tools to monitor our transportation system's physical assets and improve safety outcomes. ITS America members are working on projects across the country, including with the state of Hawaii and the Port Authority of New York and New Jersey (PANYNJ), to monitor transportation assets and inspect roadway conditions. The technology solution provides crowdsource imagery from dashcams on roadways along with AI-powered machine learning models to deliver thousands of road images per mile and automatically survey roadways for multiple different assets. These surveys included information on striping visibility, guardrail detection damage, debris, calculating PASER (Pavement Surface Evaluation and Rating) scores, and cracking. In the PANYNJ example, using the AI analysis led to an initial 25% annual savings compared to traditional inventory and manual inspection procedures.²

b) Traffic Management and Road Safety

Current approaches to traffic data collection utilize outdated technologies that are often unsafe, costly, unscalable, resource-straining, and single-purpose. AI can help avoid accidents and deaths of vulnerable roadway workers. Using artificial intelligence and high-resolution video streams, it fully automates the capture of comprehensive traffic and vehicle analytics across a variety of use cases. AI models can predict short-term traffic flows, show incident hot-spots, and predict incident delay times – giving travelers and

² <https://www.blyncsy.com/active-projects/port-authority-of-new-york-and-new-jersey/#methodology>

transportation agencies more information at their fingertips to make smarter, safer decisions.

More than 50% of the combined total of fatal and injury crashes occur at or near intersections, according to the Federal Highway Administration (FHWA). AI can be used for advanced intersection analytics and vulnerable road user (VRU) safety to be proactive, rather than the current method which is reactive. AI can analyze red-light-running data to change signal timing for pedestrians or use cameras/sensors on the roadway to automate walk/stop signal timing to allow for more pedestrians to cross the street safely. AI-driven object recognition and computer vision systems detect pedestrians, even in low-light conditions. The California Department of Transportation has begun a pilot project using GenAI for VRU safety, looking at how GenAI can transform traffic crash data into insights that help the agency proactively make safety enhancements – instead of responding to a crash when it is too late. While localized, this type of project can and should be scaled across the country with greater Federal leadership and more systematic use of technology in our transportation system.

Optimizing traffic flow at intersections has been shown to reduce congestion and lower travel times and lower crash rates. AI applications can send data to traffic management systems to adjust signal timing and to individual users to reroute cars or notify about crashes/incidents. It can take up to 70 manual hours to retime a singular signalized intersection – however there are AI tools today that can do it almost instantaneously – saving agencies time and resources.³

The Yakama Nation in the state of Washington has installed AI-powered roadside units at an intersection where the highway meets a local road, as part of a pilot project to improve traffic safety on U.S. Highway 97. These devices are equipped with multi-sensing (i.e., camera, environment sensors, etc.), computing, and communication capabilities, making them ideal for monitoring traffic, detecting dangerous events, and providing real-time warning messages to road users. These devices can operate without relying on extensive infrastructure support, such as a broadband connection. Small infrastructure upgrades such as this can have tremendous impacts on rural communities with vulnerable populations who often do not have access to broadband connections.

c) Emergency Response

Decreasing the number of traffic crashes and improving police officer response times to emergencies are critical to the health and well-being of communities across the U.S. Using cloud-based software and AI, public safety agencies can share traffic data and real-time emergency updates, helping police and emergency services reach destinations faster. Monitoring traffic patterns, real-time crashes, and road closures not only speed up response times, but it prevents more crashes from occurring.

Using cameras and other detection equipment, paired with machine learning, transportation agencies can predict traffic incident hotspots, detect crashes, and automatically signal an EMS response to dispatchers and operations. After deploying AI systems in their network, agencies have seen an upwards of 29% decrease in secondary crashes on certain highway segments due to faster notification and clearing of prior incidents. In some instances, agencies may see AI-powered insights on their roads' problem areas that

³ <https://www.flowlabs.ai/solutions/traffic-signal-operations>

they have never seen before or did not even know was an issue

Communities in Nevada and Texas, for example, are using CCTV cameras paired with AI tools to detect anomalies on highway corridors and immediately report crashes and incidents to authorities, as opposed to waiting for someone to dial in 9-1-1 and relying on often inaccurate location descriptions from callers. Agencies are experiencing an up to 23-minute reduction in EMS response time from using these tools, which is critical in the post-crash care phase of emergency response. Additionally, agencies can better position public safety personnel based on hotspots identified through AI and their crash data, allowing personnel to reach incidents of all kinds more quickly.

d) Infrastructure Planning and Design

The public and private sector are using AI to improve the way we design and plan for infrastructure upgrades, or completely new infrastructure projects. By utilizing a mix of data from information technology (IT), operational technology (OT), and engineering technology (ET) with real-life developments in a simulated environment, engineers can more effectively predict activity to mitigate future issues. In this process, AI can play a key role in building the environment in the digital realm while machine learning can be implemented to continuously predict real-life actions and their impacts on future assets.

The Minnesota Department of Transportation is using an advanced digital twinning and AI solution to improve bridge inspections across the state, with projected savings of \$4 million per year which can be used to fund other critical infrastructure improvements.⁴ With over 600,000 bridges in the United States, the opportunities are endless for using AI to improve the way we plan, design, inspect, and repair our hard infrastructure assets.

AI enables engineers to enhance efficiency, reduce errors, and quickly rework site designs. AI-powered design layout agents can evaluate thousands of layout options and suggest alternative designs in real-time, helping engineers make better design decisions sooner, saving time and money. AI in planning and design also aids in the automation of repetitive tasks like documentation, annotation, and mundane drawings to allow engineers to focus on higher-value work.

e) Vehicle Technology

Cars equipped with ADAS use versions of AI and machine learning to provide features including forward collision warning, lane departure warning, pedestrian detection and blind spot monitoring to improve safety for human drivers. ADS Level 4 and 5 in what the public generally knows as fully automated vehicles (i.e., robotaxis) are meant to operate without human input, designed to strictly obey traffic laws, follow speed limits, and come to complete stops at red lights or stop signs.

AI is used to simulate real-world conditions to safety-test AVs, and to ensure that the technology and associated programming is secure. AVs and their associated algorithms use AI to analyze road

⁴ <https://partner.microsoft.com/en-US/case-studies/bentley-systems>

conditions, make decisions while driving, and collect data while on the road.

Autonomy has great potential for the freight industry as well, and further deployment of automated trucks is expected to lower the rate of crashes and injuries on our highways involving heavy-duty trucks. In 2021, there were close to 5,800 fatalities from large truck crashes in the U.S., a number that can be lowered through AV freight technologies. Automated trucking can help alleviate distracted or drowsy driving, especially when considering the long hours drivers spend on the road. Additionally, automated freight technologies extend beyond the driver's seat, including applications that can assist with freight logistics, cargo safety, and predictive maintenance.

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Further, ITS America recommends that policymakers dedicate additional support for building greater professional cybersecurity workforce capacity within the transportation sector in order that transportation agencies have a sufficient workforce capable of preventing and responding to cybersecurity incidents.

III. Addressing AI for Automated Vehicle (AV) Technology

The AI Action Plan should consider AVs and advanced driving technology that improves road user safety and access to mobility options. The AI Action Plan from OSTP should call for the creation of such a framework and should call on USDOT to clarify and codify interpretations to ensure industry certainty on technologies and human interaction with AVs. AVs hold tremendous promise for improving roadway safety and increasing mobility options for communities of all kinds. Along with ADS Level 4 and 5, Advanced Driver Assistance Systems (ADAS) have the potential to significantly reduce crashes on our roadways. These vehicles rely on AI-powered algorithms and decision-making on the road. ITS America supports policies and regulatory frameworks that facilitate the safe testing, deployment, and integration of these technologies into the transportation system while simultaneously developing strategies to support our existing and future workforce.

In addition, **the National Highway Traffic Safety Administration (NHTSA) and Federal Motor Carrier Safety Administration (FMSCA) should finalize a comprehensive AV safety framework** expeditiously so that we can fully realize the benefits of automation on our roads on a national level.

IV. Promoting Explainability, Transparency, and Trust

The OSTP AI Action Plan should direct USDOT to develop transparency guidelines for AI applications in coordination with the industry. Bringing AI innovation to transportation, a sector where safety and consumer acceptance is paramount, will require AI systems to be explainable, transparent, and trustworthy. Without transparency, advanced AI can be susceptible to unethical uses and reduce public trust. Establishing a

¹ ¹ National Institute of Standards and Technology. <https://www.nist.gov/itl/ai-risk-management-framework>

level of transparency in data collection, algorithms, and how AI applications are used will help ensure trust among users. Transparency about data and AI models encourages responsibility, decreases the risk of misuse, and will ultimately lead to better tools and better results for the American public. USDOT and other Federal stakeholders should develop robust reporting requirements for any harmful incidents associated with AI utilization in the name of greater transparency.

Additionally, **ITS America recommends the development of explainability standards in the Action Plan for the use of AI systems in transportation, as well as other industries.** AI tools should be explainable to those without expertise in technology or a specific industry. Without strong explainability, it will be more difficult to improve consumer adoption of technologies, will further erode trust, and may be prohibitive to future private sector investment. If there is a lack of trust among consumers, private companies may not be incentivized to innovate and create new products and services in the AI realm.

The Administration's Action Plan should ensure that the AI Safety Institute continues its work in creating safety standards, testing new technologies, and evaluating AI models. Without such a body, the U.S. risks ceding AI leadership to non-allied countries such as China and opening the door to the proliferation of unsafe and untrustworthy AI models. The AI Safety Institute has broad industry support beyond the transportation sector, as well as bipartisan support from Congress.

V. Strengthening Workforce Development and Education

ITS America recommends that the Administration, in coordination with Federal agencies such as the Department of Labor and USDOT, develop a robust workforce development plan and assistance program to ensure our transportation workforce is prepared to leverage AI tools and solutions. In the era of AI, workforce development and education are critical to fostering broad adoption and ensuring a pipeline of innovation. In any industry, upskilling and reskilling may be necessary to fully maximize the benefits of AI applications. This includes training the existing workforce on how to leverage AI tools and solutions to support their own work, in addition to AI's capabilities and limitations. AI should be used for decision support and augmentation of tasks, not as a replacement, and workforce development programs should reflect that.

In the AI Action Plan, OSTP should place emphasis on the need to conduct AI training for practitioners and those who will be interacting with and using AI on the job. Federal leaders can and should put the transportation workforce ahead of the curve on AI. As AI adoption grows in the transportation industry, we must ensure that Federal transportation workforce programs are prioritized and well-funded for digital skills training.

For future workforce needs, OSTP and USDOT should continue to support the development of AI training curriculum related to transportation at universities and should prioritize AI as a key component of the University Transportation Centers program. Additionally, AI literacy and curriculum should be a core component of community colleges, apprenticeships, career and technical education, and even primary and secondary education as we look to prepare our workforce for the future.

VI. Modernizing Procurement Practices

The AI Action Plan should discuss ways to streamline the contracting process for technology solutions, such as AI, and OSTP should work with USDOT to provide practices for procuring AI-based tools for transportation projects. This may include continued research and pilots on innovative procurement methods through ARPA-I or FHWA's Every Day Counts program. Procurement remains a core barrier to AI adoption in transportation. Procurement policies designed for physical infrastructure do not adequately consider the procurement, maintenance, and operations needs of a technology-inclusive infrastructure system. Data streams, software licenses, and even cloud storage services are key to the transportation system today and the use of AI but were not core parts of our infrastructure procurement thirty years ago. AI solutions are distinctly unique from physical assets – they are not one-time purchases, and they often require ongoing upgrades to ensure cybersecurity and data integrity and can improve performance over time unlike a static physical asset.

Procurement on Federally-funded transportation projects should allow for and encourage license-based or subscription products that are often associated with technologies like AI – these are constantly updating assets and not one-time hard purchases. The current transportation technology procurement process can be lengthy and onerous, especially for States and localities that may lack the necessary resources and expertise, slowing down projects and threatening the efficiency of investment in technology solutions. Additionally, Federal regulations pertaining to operational costs in Federal-aid Highway projects make it difficult to fund the ongoing operations and maintenance of assets. AI assets require ongoing maintenance and optimization to work effectively.

Conclusion

To solve our safety crisis on the road, make travel more efficient, and spur economic growth, we must take bold steps toward transforming how we integrate transportation technologies, including AI, into our transportation system at scale. AI capabilities are advancing at a fast pace, and we need a Federal policy environment that allows us to more holistically integrate these tools today so that we are prepared to take advantage of the new and improved capabilities of tomorrow. Doing things the old way has resulted in less safe roadways, more congested commutes, slower movements of goods and broken supply chains, and decreasing global competitiveness. Now is the time to accelerate our investments in AI and empower our transportation system for a better tomorrow. We look forward to seeing significant consideration of transportation in the Administration's AI Action Plan.

If you have any questions, please contact Bobby McCurdy, Vice President of Policy & Advocacy at

Laura Chace
President and CEO
Intelligent Transportation Society of America

Appendix: Applications

a) Proactive Maintenance (Infrastructure and Vehicles) and Asset Management

Using AI, transportation agencies can predict maintenance needs proactively for infrastructure assets and fleets. AI tools can help agencies identify structural issues in bridges and roads early on and monitor for further wear-and-tear. Traditional methods involve manual inspection, which can be costly and time consuming for public agencies. New innovations use computer vision AI, virtual reality tools, and drones to collect images and videos of roads and bridges. While the cameras continually monitor the site, software processes and analyzes the collected data, providing engineers with a safety assessment that includes information about structural changes and weaknesses, as well as immediate damage.

Using machine learning and other AI tools, automakers and manufacturers can now identify and predict vehicle and fleet maintenance to a more precise degree, improving not only vehicle performance but also maintenance operations and costs. New York City's MTA is using AI technology to predict bus breakdowns and maintenance needs, helping the agency plan for future needs and operate efficiently for customers. MTA found a 75% increase in maintenance productivity and material costs lowered by 24%.

Maintenance through software-enabled assets is a key area where USDOT and the AI Action Plan could focus, bringing more cutting-edge AI tools to monitor our transportation system's physical assets and improve safety outcomes. ITS America members are working on projects across the country, including with the state of Hawaii and the Port Authority of New York and New Jersey (PANYNJ), to monitor transportation assets and inspect roadway conditions. The technology solution provides crowdsource imagery from dashcams on roadways along with AI-powered machine learning models to deliver thousands of road images per mile and automatically survey roadways for multiple different assets. These surveys included information on striping visibility, guardrail detection damage, debris, calculating PASER (Pavement Surface Evaluation and Rating) scores, and cracking. In the PANYNJ example, using the AI analysis led to an initial 25% annual savings compared to traditional inventory and manual inspection procedures.²

b) Traffic Management and Road Safety

Current approaches to traffic data collection utilize outdated technologies that are often unsafe, costly, unscalable, resource-straining, and single-purpose. AI can help avoid accidents and deaths of vulnerable roadway workers. Using artificial intelligence and high-resolution video streams, it fully automates the capture of comprehensive traffic and vehicle analytics across a variety of use cases. AI models can predict short-term traffic flows, show incident hot-spots, and predict incident delay times – giving travelers and

² <https://www.blyncsy.com/active-projects/port-authority-of-new-york-and-new-jersey/#methodology>

transportation agencies more information at their fingertips to make smarter, safer decisions.

More than 50% of the combined total of fatal and injury crashes occur at or near intersections, according to the Federal Highway Administration (FHWA). AI can be used for advanced intersection analytics and vulnerable road user (VRU) safety to be proactive, rather than the current method which is reactive. AI can analyze red-light-running data to change signal timing for pedestrians or use cameras/sensors on the roadway to automate walk/stop signal timing to allow for more pedestrians to cross the street safely. AI-driven object recognition and computer vision systems detect pedestrians, even in low-light conditions. The California Department of Transportation has begun a pilot project using GenAI for VRU safety, looking at how GenAI can transform traffic crash data into insights that help the agency proactively make safety enhancements – instead of responding to a crash when it is too late. While localized, this type of project can and should be scaled across the country with greater Federal leadership and more systematic use of technology in our transportation system.

Optimizing traffic flow at intersections has been shown to reduce congestion and lower travel times and lower crash rates. AI applications can send data to traffic management systems to adjust signal timing and to individual users to reroute cars or notify about crashes/incidents. It can take up to 70 manual hours to retime a singular signalized intersection – however there are AI tools today that can do it almost instantaneously – saving agencies time and resources.³

The Yakama Nation in the state of Washington has installed AI-powered roadside units at an intersection where the highway meets a local road, as part of a pilot project to improve traffic safety on U.S. Highway 97. These devices are equipped with multi-sensing (i.e., camera, environment sensors, etc.), computing, and communication capabilities, making them ideal for monitoring traffic, detecting dangerous events, and providing real-time warning messages to road users. These devices can operate without relying on extensive infrastructure support, such as a broadband connection. Small infrastructure upgrades such as this can have tremendous impacts on rural communities with vulnerable populations who often do not have access to broadband connections.

c) Emergency Response

Decreasing the number of traffic crashes and improving police officer response times to emergencies are critical to the health and well-being of communities across the U.S. Using cloud-based software and AI, public safety agencies can share traffic data and real-time emergency updates, helping police and emergency services reach destinations faster. Monitoring traffic patterns, real-time crashes, and road closures not only speed up response times, but it prevents more crashes from occurring.

Using cameras and other detection equipment, paired with machine learning, transportation agencies can predict traffic incident hotspots, detect crashes, and automatically signal an EMS response to dispatchers and operations. After deploying AI systems in their network, agencies have seen an upwards of 29% decrease in secondary crashes on certain highway segments due to faster notification and clearing of prior incidents. In some instances, agencies may see AI-powered insights on their roads' problem areas that

³ <https://www.flowlabs.ai/solutions/traffic-signal-operations>

they have never seen before or did not even know was an issue

Communities in Nevada and Texas, for example, are using CCTV cameras paired with AI tools to detect anomalies on highway corridors and immediately report crashes and incidents to authorities, as opposed to waiting for someone to dial in 9-1-1 and relying on often inaccurate location descriptions from callers. Agencies are experiencing an up to 23-minute reduction in EMS response time from using these tools, which is critical in the post-crash care phase of emergency response. Additionally, agencies can better position public safety personnel based on hotspots identified through AI and their crash data, allowing personnel to reach incidents of all kinds more quickly.

d) Infrastructure Planning and Design

The public and private sector are using AI to improve the way we design and plan for infrastructure upgrades, or completely new infrastructure projects. By utilizing a mix of data from information technology (IT), operational technology (OT), and engineering technology (ET) with real-life developments in a simulated environment, engineers can more effectively predict activity to mitigate future issues. In this process, AI can play a key role in building the environment in the digital realm while machine learning can be implemented to continuously predict real-life actions and their impacts on future assets.

The Minnesota Department of Transportation is using an advanced digital twinning and AI solution to improve bridge inspections across the state, with projected savings of \$4 million per year which can be used to fund other critical infrastructure improvements.⁴ With over 600,000 bridges in the United States, the opportunities are endless for using AI to improve the way we plan, design, inspect, and repair our hard infrastructure assets.

AI enables engineers to enhance efficiency, reduce errors, and quickly rework site designs. AI-powered design layout agents can evaluate thousands of layout options and suggest alternative designs in real-time, helping engineers make better design decisions sooner, saving time and money. AI in planning and design also aids in the automation of repetitive tasks like documentation, annotation, and mundane drawings to allow engineers to focus on higher-value work.

e) Vehicle Technology

Cars equipped with ADAS use versions of AI and machine learning to provide features including forward collision warning, lane departure warning, pedestrian detection and blind spot monitoring to improve safety for human drivers. ADS Level 4 and 5 in what the public generally knows as fully automated vehicles (i.e., robotaxis) are meant to operate without human input, designed to strictly obey traffic laws, follow speed limits, and come to complete stops at red lights or stop signs.

AI is used to simulate real-world conditions to safety-test AVs, and to ensure that the technology and associated programming is secure. AVs and their associated algorithms use AI to analyze road

⁴ <https://partner.microsoft.com/en-US/case-studies/bentley-systems>

conditions, make decisions while driving, and collect data while on the road.

Autonomy has great potential for the freight industry as well, and further deployment of automated trucks is expected to lower the rate of crashes and injuries on our highways involving heavy-duty trucks. In 2021, there were close to 5,800 fatalities from large truck crashes in the U.S., a number that can be lowered through AV freight technologies. Automated trucking can help alleviate distracted or drowsy driving, especially when considering the long hours drivers spend on the road. Additionally, automated freight technologies extend beyond the driver's seat, including applications that can assist with freight logistics, cargo safety, and predictive maintenance.